Project
PRISM

Recombination

Genetics and Evolution

What is Recombination?

- **Genetic recombination** is the production of offspring with allele combinations different from those found in either parental chromosome arrangement.
- It creates **new combinations of existing alleles**.
- Recombination increases genetic diversity without creating new alleles.
- New alleles arise by **mutation**; recombination **reshuffles** alleles already present.

Two Sources of Recombination

- Recombination can arise in two different ways:

  independent assortment of genes on different chromosomes

  crossing over between linked genes on homologous chromosomes

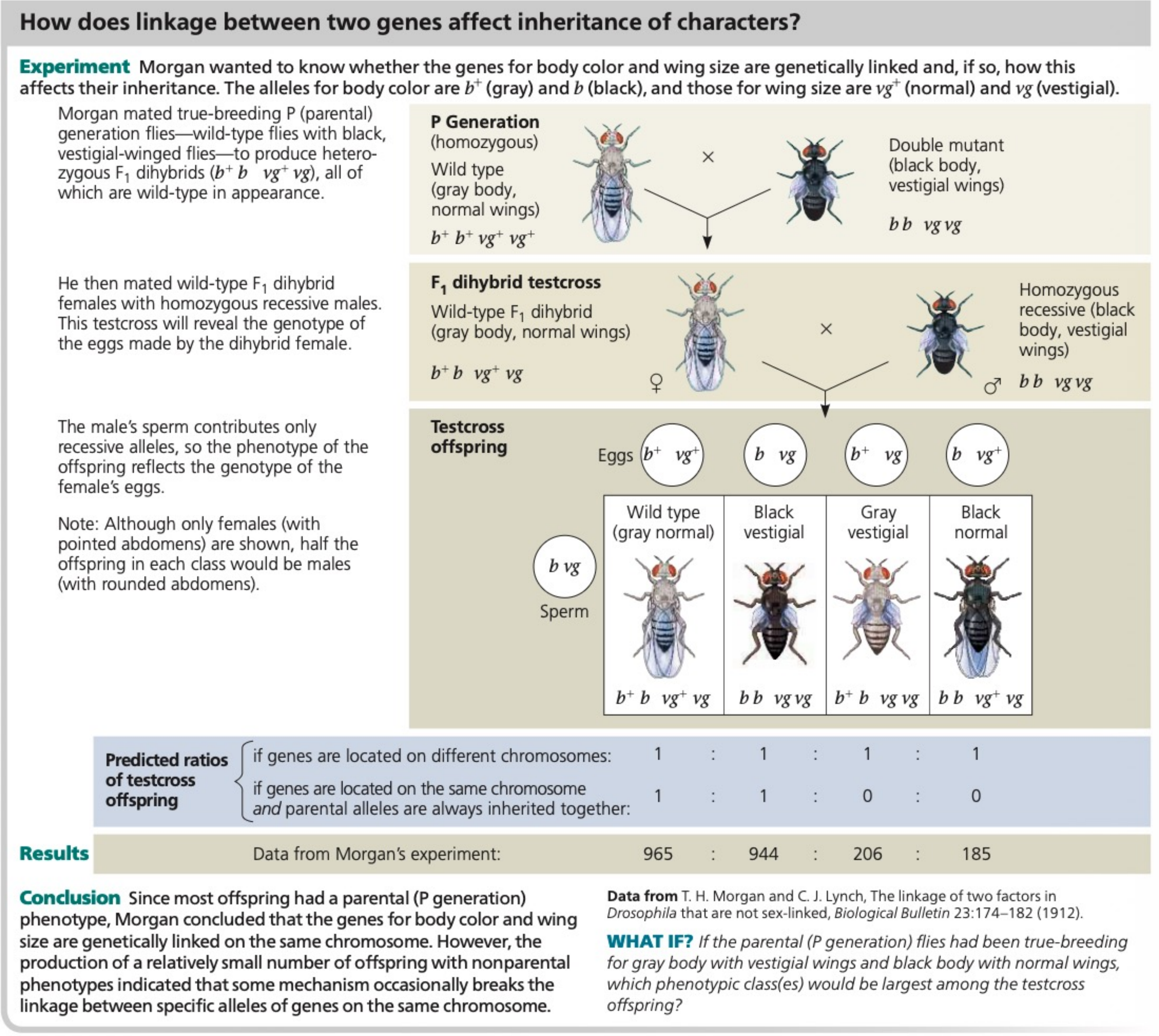
- Therefore, recombination is a broader idea than crossing over.
- Crossing over is only one mechanism that can produce recombinant genotypes.

Recombination of Unlinked Genes

- If two genes are on **different chromosomes**, they assort independently during meiosis.
- In this case, recombinant offspring are produced by **independent assortment**.
- This is why a dihybrid testcross with unlinked genes gives the expected **1:1:1:1** ratio.
- For unlinked genes, recombinant and parental combinations are expected in equal frequency.

Linked Genes

- **Linked genes** are genes located on the same chromosome.
- Because they are physically connected, they tend to be inherited together.
- Therefore, they do **not** usually show the same ratios expected for independently assorting genes.
- However, linked genes are not inherited together all the time, because crossing over can separate them.



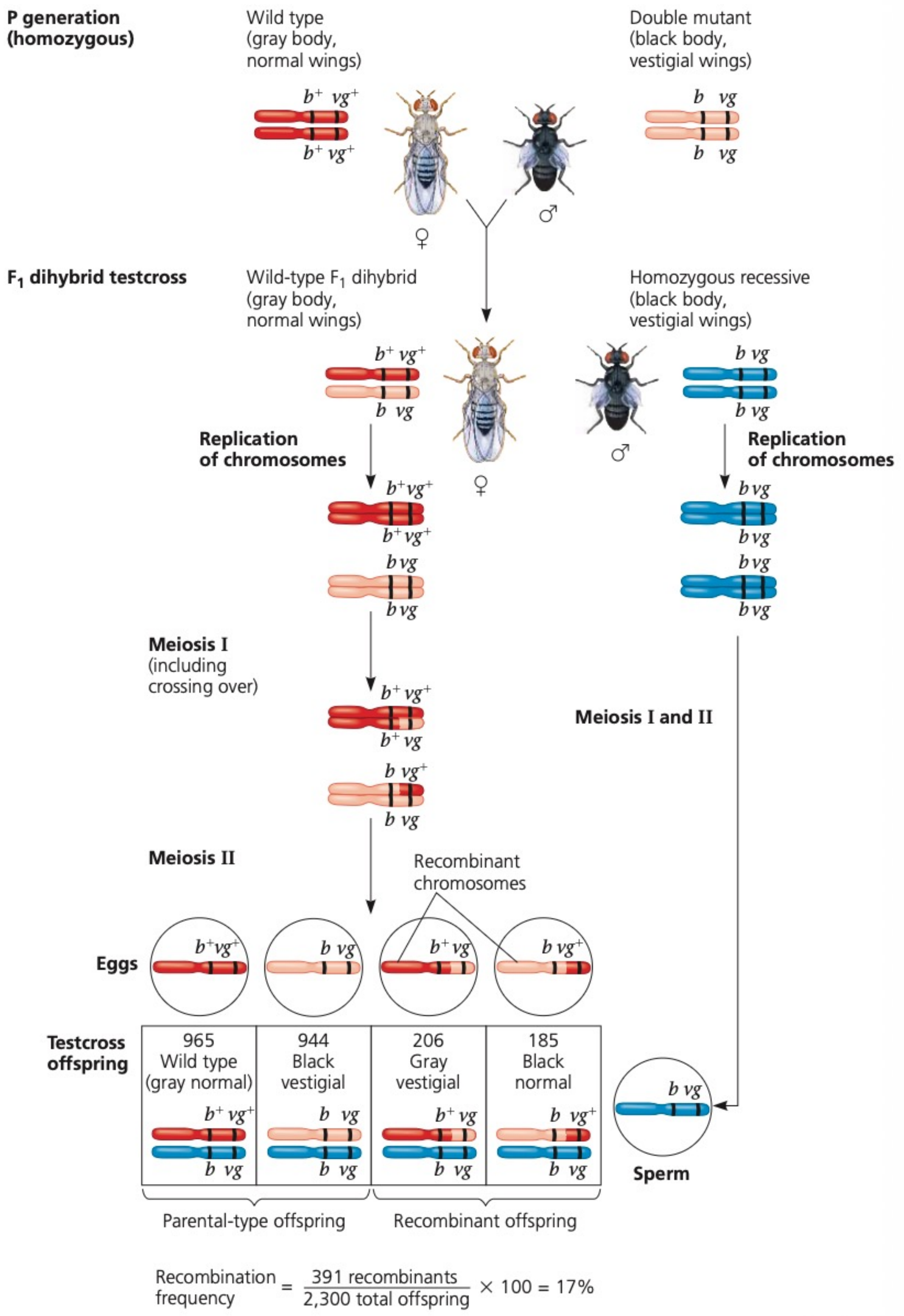
Parental Types and Recomb

- In linkage problems, offspring are divided into:

  parental types — allele combinations already present in the original parental chromosomes

  recombinant types — new allele combinations produced by recombination

- For linked genes, parental types usually occur more often than recombinant types.
- This is one of the main signs that the genes are linked.



Coupling and Repulsion

- In a heterozygote, linked alleles can be arranged in two ways:

coupling (cis): both dominant alleles are on one homologue and both recessive alleles are on the other

example: **AB / ab**

repulsion (trans): each homologue carries one dominant and one recessive allele

example: **Ab / aB**

- These arrangements determine which offspring classes are **parental** and which are **recombinant.**
- Therefore, before identifying recombinant classes, the parental chromosome arrangement must first be determined.

Crossing Over

- During **prophase I of meiosis**, homologous chromosomes pair.
- Nonsister chromatids may exchange corresponding DNA segments.
- This process is called **crossing over**.
- Crossing over produces **recombinant chromosomes**.
- These recombinant chromosomes can then enter gametes and generate recombinant offspring.

Chiasmata and Physical Basis

- Crossing over occurs after homologous chromosomes pair and before they separate.
- The visible crossover points between homologous chromosomes are called **chiasmata**.
- Recombination therefore has a direct **chromosomal basis**, not just a statistical one.
- This is why linkage and recombination connect chromosome behaviour in meiosis to inheritance patterns.

Testcrosses Reveal Recombination

- Recombination is usually measured with a **testcross**.
- A heterozygote for two genes is crossed with a **double recessive** individual.
- Because the recessive parent contributes only recessive alleles, each offspring phenotype reveals the gamete produced by the heterozygous parent.
- This makes it possible to count **parental** and **recombinant** classes directly.

Recombination Frequency

- The **recombination frequency** measures how often recombinant offspring are produced.
- It is calculated as:

$$Recombination\ frequency\ (\%) = \frac{number\ of\ recombinant\ offsprings}{total\ number\ of\ offsprings} \times 100\%$$

- If recombination frequency is **less than 50%**, the genes are genetically linked.
- The lower the recombination frequency, the closer the genes are likely to be.

The 50% Limit

- Recombination frequency can never exceed **50%**.
- If two genes show **50% recombination**, they behave as if they are unlinked.
- This may happen because:

  they are on different chromosomes, or

  they are very far apart on the same chromosome
- So **50% recombination does not prove different chromosomes**; it only means linkage is not detectable from the cross.

Linkage Maps

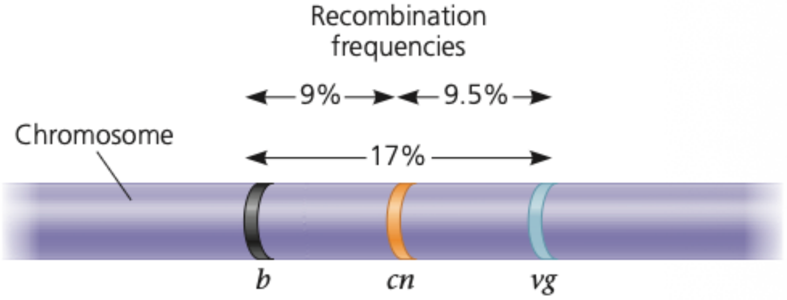
- Recombination frequency can be used to estimate the relative positions of genes.
- A genetic map based on recombination data is called a **linkage map**.
- **1 map unit = 1% recombination frequency**
- **1 map unit = 1 centimorgan (cM)**
- Genes are arranged in the order that best fits the recombination data.

Constructing a Linkage Map

Application A linkage map shows the relative locations of genes along a chromosome.

Technique A linkage map is based on the assumption that the probability of a crossover between two genetic loci is proportional to the distance separating the loci. The recombination frequencies used to construct a linkage map for a particular chromosome are obtained from experimental crosses, such as the cross depicted in Figures 15.9 and 15.10. The distances between genes are expressed as map units, with one map unit equivalent to a 1% recombination frequency. Genes are arranged on the chromosome in the order that best fits the data.

Results In this example, the observed recombination frequencies between three *Drosophila* gene pairs (*b* and *cn*, 9%; *cn* and *vg*, 9.5%; and *b* and *vg*, 17%) best fit a linear order in which *cn* is positioned about halfway between the other two genes:



The *b*-*vg* recombination frequency (17%) is slightly less than the sum of the *b*-*cn* and *cn*-*vg* frequencies (9 + 9.5 = 18.5%) because of the few times that one crossover occurs between *b* and *cn* and another crossover occurs between *cn* and *vg*. The second crossover would "cancel out" the first, reducing the observed *b*-*vg* recombination frequency while contributing to the frequency between each of the closer pairs of genes. The value of 18.5% (18.5 map units) is closer to the actual distance between the genes. In practice, a geneticist would add the smaller distances in constructing a map.

Why Distance Matters

- The farther apart two genes are, the more possible crossover points lie between them.
- Therefore, genes farther apart usually show a **higher recombination frequency**.
- This is why recombination frequency can be used as an estimate of relative distance.
- Sturtevant used this principle to build some of the first genetic maps.

Why Maps Are Approximate

- Linkage maps show **gene order** reliably, but not exact physical DNA distance.
- One reason is that **double crossovers** can occur.
- A second crossover can restore the original outside marker arrangement and therefore hide part of the real crossover history.
- As a result, observed recombination frequency may **underestimate true distance** between genes.

Three-Point Crosses

- A **three-point testcross** allows geneticists to determine:

  gene order

  distances between adjacent genes

  whether double crossovers occurred
- The rarest offspring classes are often the **double crossover** classes.
- Comparing these with parental classes helps identify the **middle gene**.

Coefficient of Coincidence and Interference

- If double crossovers occurred exactly as often as expected from independent probabilities, there would be **no interference**.
- In reality, one crossover often reduces the chance of another nearby crossover.
- This is measured using:

  coefficient of coincidence

  interference
- **Interference** is common in gene-mapping problems and is important!

Tetrad Analysis and Centromere Mapping

- In some fungi, the products of a single meiosis remain together in an ordered structure called a **linear tetrad**.
- This allows recombination events to be studied more directly than in ordinary offspring counts.
- Tetrad analysis can be used to infer:

  linkage

  double crossovers

  gene-centromere distance

Biological Importance of Recombination

- Recombination creates new allele combinations in populations.
- This increases the diversity on which **natural selection** can act.
- It is therefore important both genetically and evolutionarily.
- Recombination links meiosis directly to population-level variation and adaptation.

New Combinations of Alleles: Variation for Natural Selection

EVOLUTION The physical behavior of chromosomes during meiosis contributes to the generation of variation in offspring (see Concept 13.4). Each pair of homologous chromosomes lines up independently of other pairs during metaphase I, and crossing over prior to that, during prophase I, can mix and match parts of maternal and paternal homologs. Mendel's elegant experiments show that the behavior of the abstract entities known as genes—or, more concretely, alleles of genes—also leads to variation in offspring (see Concept 14.1). Now, putting these different ideas together, you can see that the recombinant chromosomes resulting from crossing over may bring alleles together in new combinations, and the subsequent events of meiosis distribute to gametes the recombinant chromosomes in a multitude of combinations, such as the new variants seen in Figures 15.9 and 15.10. Random fertilization then increases even further the number of variant allele combinations that can be created.

This abundance of genetic variation provides the raw material on which natural selection works. If the traits conferred by particular combinations of alleles are better suited for a given environment, organisms possessing those genotypes will be expected to thrive and leave more offspring, ensuring the continuation of their genetic complement. In the next generation, of course, the alleles will be shuffled anew. Ultimately, the interplay between environment and phenotype (and thus genotype) will determine which genetic combinations persist over time.

Conclusion

- Recombination produces new allele combinations.
- For **unlinked genes**, recombination results from independent assortment.
- For **linked genes**, recombination results mainly from crossing over during meiosis I.
- Recombination frequency allows estimation of gene order and distance.
- Advanced analysis includes **three-point crosses**, **double crossovers**, **interference**, and **tetrad analysis**.
- Recombination is a major source of heritable variation in sexually reproducing organisms

IBO past year questions

IBO 2016 Theoretical Exam B: Qn 84

An organism has four genes, A, B, C and D with two alleles each. An individual heterozygous for these genes was bred with one that is homozygous recessive. The cross produced 3288 offspring with phenotypes shown in the table below:

Phenotypes	Number of individuals
ABCD	675
ABCd	83
ABcD	1
ABcd	74
AbCD	73
AbCd	1
AbcD	84
Abcd	670
aBCD	655
aBCd	86
aBcD	1
aBcd	73
abCD	71
abCd	1
abcD	87
abcd	653

Indicate in the answer sheet if each of the following statements is true or false.

A. The four loci are genetically linked.

B. The distance between gene B and gene D is 9 cM.

C. The distance between gene D and gene C is 10.5 cM.

D. Interference happened with a value less than 0.25. Interference = 1-(observed frequency of double crossover/expected frequency of double crossover).

*IBO 2013 Theoretical Exam 2: Qn 30

In a plant species, the level of anthocyanin pigments produced is controlled by a single gene G, for which only a "dark" and a "light" allele are present. To more accurately map the position of gene G on chromosome 3, two inbred lines (P1 and P2) are crossed and F2 individuals (X1 through X5) are genotyped at five single nucleotide variant loci (SNV1 through SNV5) on the same chromosome.

	SNV1	SNV2	SNV3	SNV4	SNV5	Anthocyanin [mM]
P1	A/A	C/C	A/A	T/T	C/C	120
P2	T/T	T/T	C/C	G/G	G/G	25
X1	A/T	C/T	A/C	T/G	C/G	115
X2	A/A	C/C	A/A	T/G	C/G	123
X3	A/A	T/T	C/C	T/G	C/G	22
X4	A/A	C/T	C/C	T/G	C/C	29
X5	A/T	C/T	A/A	T/T	C/C	118

Based on these results, indicate if each of the following statements is true or false.

A. One recombination event happened in each parent of X2 between the genotyped loci.

B. F1 individuals are likely showing intermediate levels of anthocyanin.

C. Among the studied loci, SNV3 is closest to gene G.

D. The phenotype ratio in the progeny of a cross between X4 and X5 is 2:1.

Answer Key

Qn 84

A. False B. False C. True D. False

Explanation

A. False. Genes BCD are linked but gene A is unlinked with them.

B. False. The distance B-D: 100% (83+74+73+84+86+73+71+87)/3288 = 19 m.u.

C. True. The distance C-D: 100% (87+1+1+86+84+1+1+83)/3288 = 10.5 m.u.

D. False. Interference (I) = 1- [(observed DCO)/(expected DCO)] = 1-{[(1+1+1+1)/3288]/[(0.0897) (0.105)]} = 0.89.

Qn 30

A. False **B.** False **C.** True **D.** False

Original commentary
Correct answers
A *false*
Most likely, a single recombination event happened in only one of the parents, as one of the haplotype is parental (ACATC) and the other shows a recombination between SNV3 and SNV4.
B *false*
Gene G is linked to SNV3, and the table lists a heterozygous individual (X1) with an elevated anthocyanin level (which is the dominant allele). But note that the students do not need to identify the most closely linked locus as for each of them heterozygous individuals are given and the conclusion would remain unchanged.
C *true*
This is the only locus for which the genotypes match the phenotypes in a Mendelian fashion.
D *false*
A 2:1 ratio is impossible for any crossing, as we clearly have dominant-recessive inheritance. So it is possible to answer this question even if the truly linked locus is not identified.

End of slides